

# QA Tasks Demo - Project Context

## Objective

Build a mobile Android application in Kotlin using Jetpack Compose as a portfolio project for QA Automation.

The app should demonstrate:

- Android development fundamentals
  - MVVM architecture
  - Navigation Compose
  - State management
  - Future automation using Maestro
  - Future API testing scenarios
  - Future Room persistence testing
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## Current Progress

Completed:

- Login Screen
- Navigation setup
- Task List Screen
- Add Task Screen
- Task model
- TaskViewModel
- Floating Action Button (Add Task)

Pending:

- Edit Task
  - Mark Task as Completed
  - Delete Task
  - Pull To Refresh
  - Error Handling
  - Room Database
  - API Integration
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## Project Structure

presentation/ |—— navigation/ | |—— AppNavigation.kt |—— screens/ | |—— login/ | |—— tasks/  
| |—— TaskListScreen.kt | |—— AddTaskScreen.kt | |—— TaskViewModel.kt |—— components/ |  
|—— TaskCard.kt

domain/ |—— model/ |—— Task.kt

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## Task Model

data class Task( val id: Int, val title: String, val description: String, val completed: Boolean = false )

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## Current Navigation Flow

LoginScreen ↓ TaskListScreen ↓ AddTaskScreen ↓ Back to TaskListScreen

Routes:

- login
  - tasks
  - add\_task
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## Current ViewModel

TaskViewModel manages:

- addTask()
- deleteTask()
- toggleCompleted()

State is stored using:

MutableStateFlow<List<Task>>

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## Architecture Goal

MVVM

UI (Compose) ↓ ViewModel ↓ Repository ↓ Room (future)

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## QA Automation Goals

Future automation scenarios:

- Login validation
- Create task
- Edit task
- Delete task
- Mark task complete
- Pull to refresh
- Error handling
- Offline mode
- Room persistence

Tools planned:

- Maestro
- Playwright (web portfolio)
- Android Emulator
- GitHub Actions

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## Coding Guidelines

- Kotlin
- Jetpack Compose
- Material 3
- Navigation Compose
- StateFlow
- ViewModel
- Clean Architecture principles
- Testability first

Always propose solutions following Android best practices and maintain compatibility with modern Android SDK versions.